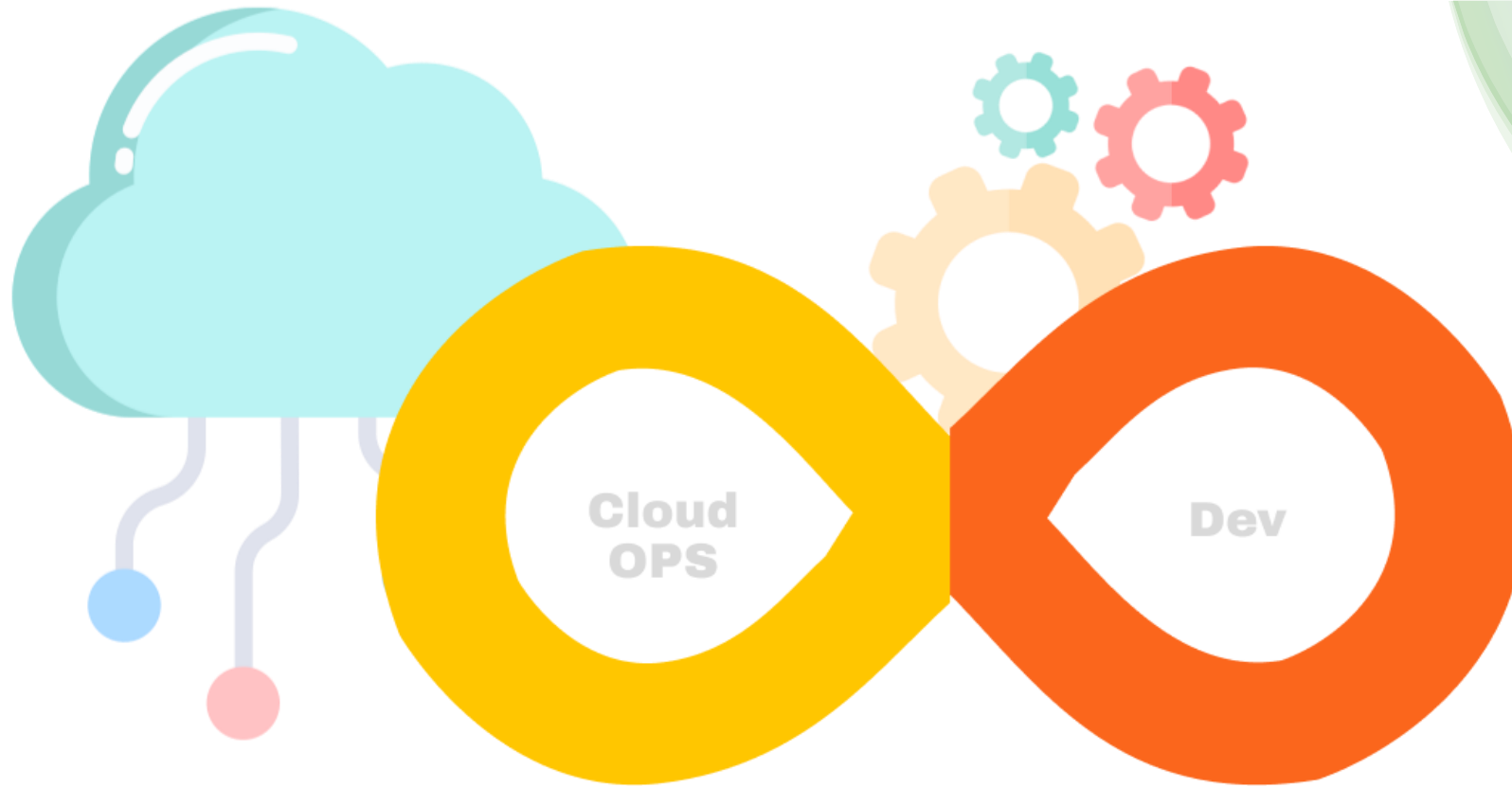




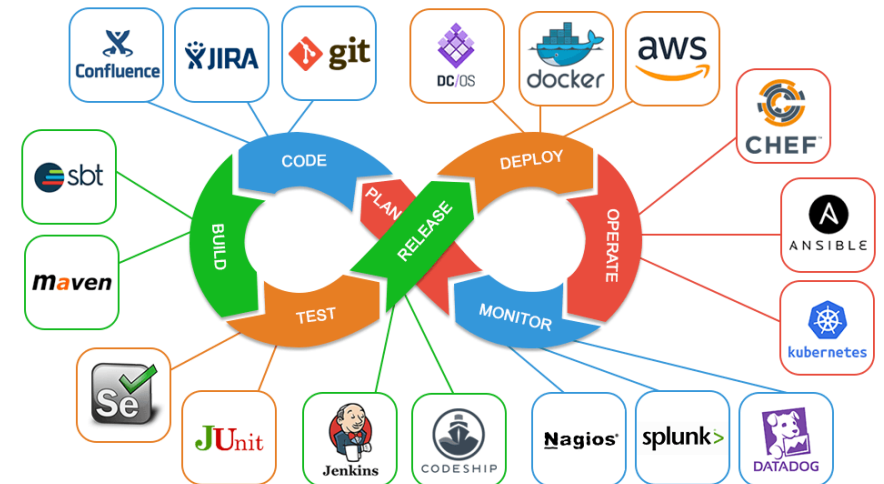
Introduction to DevOps and Cloud Computing

Muhammad Ali

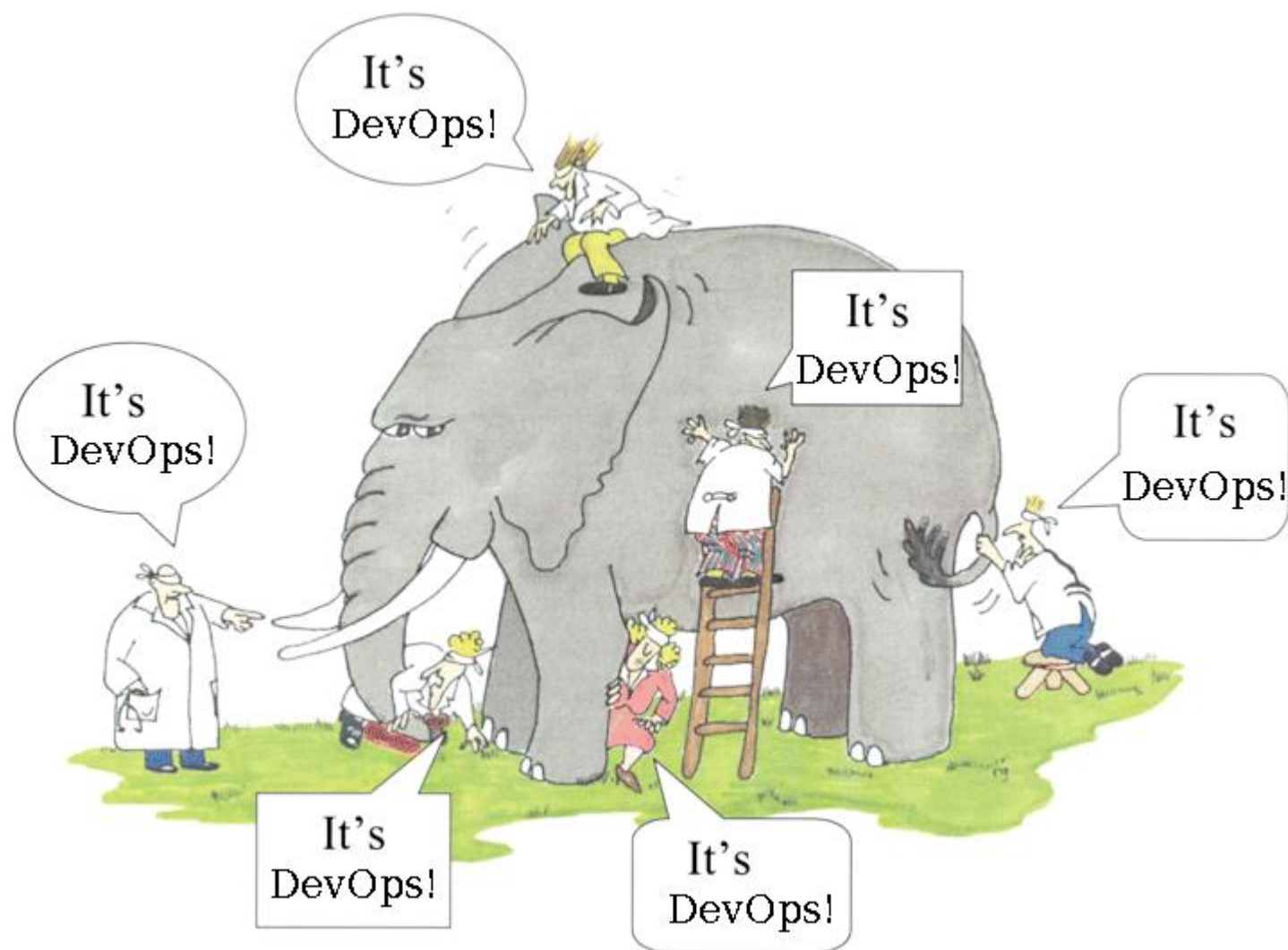
Course : DevOps and Cloud Computing Foundations
Department of Computer Science, SCIT, Beaconhouse National University
(BNU)

Agenda

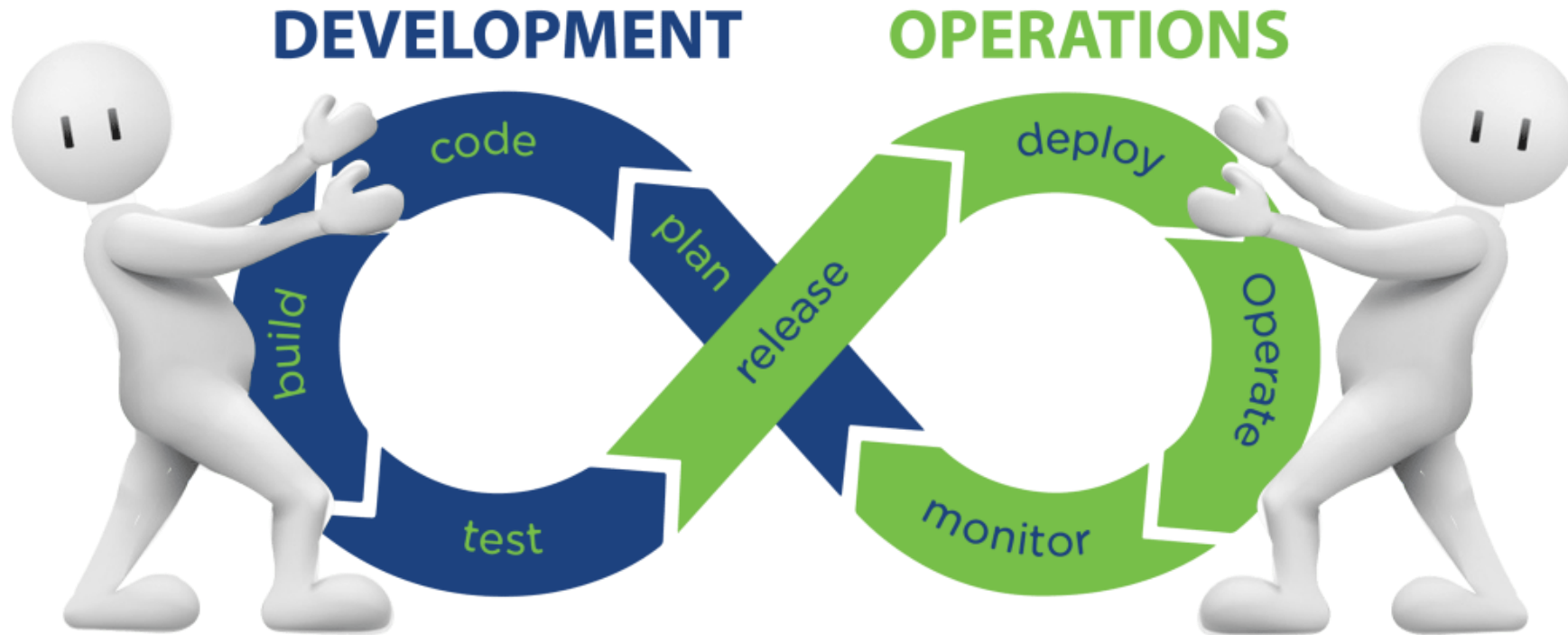
- What is DevOps?
- What is Cloud Computing?
- Core Concepts of DevOps
- DevOps Culture
- Principles of DevOps
- Overview of DevOps Lifecycle
- Relationship Between DevOps and Cloud Computing

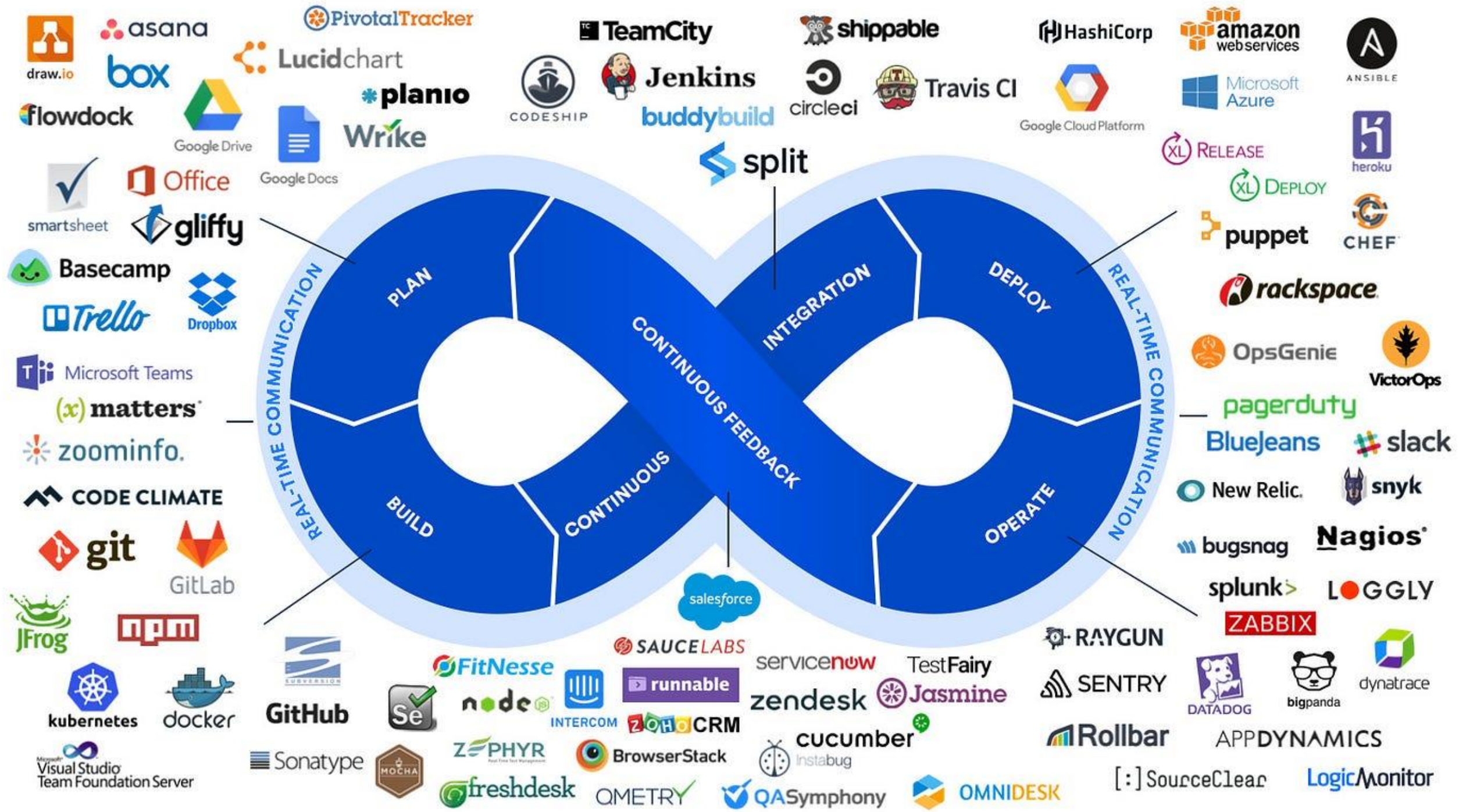


What is DevOps?



What is DevOps?





What is DevOps?

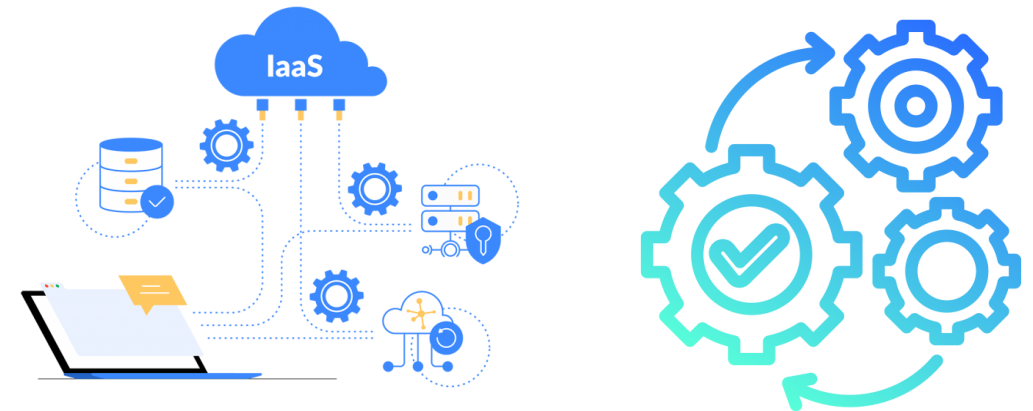
DevOps is a cultural and technical movement that aims to unify software development (Dev) and IT operations (Ops) to improve collaboration, efficiency, and automation across the software delivery pipeline.

Goals of DevOps:

- Increase deployment frequency
- Achieve faster time to market
- Improve collaboration between developers and operations
- Enhance quality and stability of software

Key Focus Areas:

- Continuous Integration (CI)
- Continuous Delivery (CD)
- Infrastructure as Code (IaC)
- Automation



What is Cloud Computing?



What is Cloud Computing?



What is Cloud Computing?

Cloud computing delivers (e.g., storage, software, etc.) over the Internet.

Key Characteristics:

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Cloud Models:

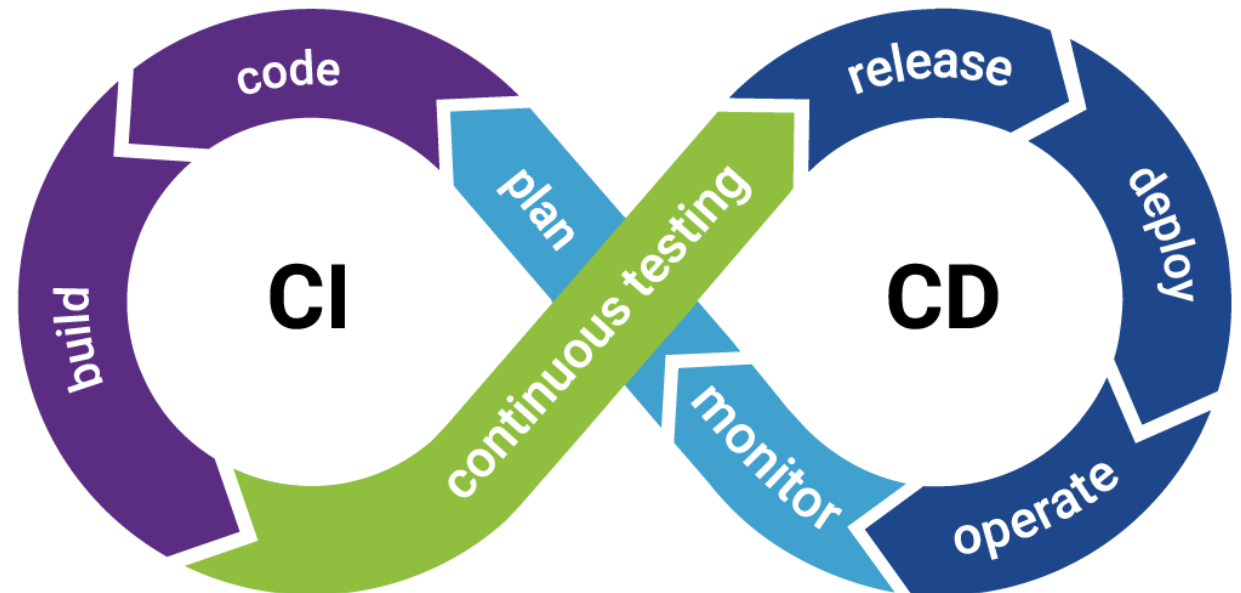
- IaaS (Infrastructure as a Service)
- PaaS (Platform as a Service)
- SaaS (Software as a Service)



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Core Concepts of DevOps

- Collaboration
- Automation
- Continuous Integration (CI)
- Continuous Delivery
- Monitoring and Feedback



DevOps Culture



Collaboration Over Silos:

Development, operations, and other teams work together to achieve common goals, fostering open communication and shared responsibility.



Shift-Left Mindset:

Moving testing and quality assurance processes earlier in the software development lifecycle to identify and resolve issues sooner.



Blameless Culture:

A focus on learning from failures rather than assigning blame. This encourages innovation and risk-taking in a supportive environment.



Embracing Automation:

Automating tasks (e.g., testing, deployment) improves speed and consistency, reducing human error and freeing up team members for more strategic work.



Continuous Improvement:

Teams regularly review and improve processes, tools, and practices to increase efficiency and effectiveness.

Overview of the DevOps Lifecycle

Development Phase:	Coding and planning Collaboration with stakeholders
Build Phase:	Code compilation, version control Continuous Integration to automate build and testing
Test Phase:	Automated testing to ensure code quality Continuous testing during integration
Release Phase:	Continuous Delivery/Deployment for faster release cycles Feature toggles, rollbacks
Deploy Phase:	Deploying to staging and production environments Infrastructure as Code (IaC) for automation
Operate Phase:	Monitoring the application in production Incident response and troubleshooting
Monitor Phase:	Continuous monitoring for performance and feedback Log management and real-time metrics

Relationship Between DevOps and Cloud Computing

DevOps and Cloud Computing Synergy:

- Cloud platforms (AWS, Azure, GCP) provide flexible, scalable infrastructure for DevOps practices.
- Cloud computing allows DevOps teams to automate resource provisioning (IaC) and implement CI/CD pipelines.
- Cloud's elasticity supports scaling of resources in response to varying loads, improving the DevOps lifecycle's agility.

Benefits of Using Cloud in DevOps:

- **Scalability:** Easily scale up or down depending on demand.
- **Availability:** Cloud services are highly available, supporting 24/7 deployment pipelines.
- **Cost Efficiency:** Pay-as-you-go models reduce upfront capital expenses for infrastructure.
- **Speed:** Faster provisioning and deployment with cloud-based resources.

